Homework #6 — Hypothesis Testing (15 pts)

MATH 240 — Introduction to Statistics, Fall 2026

Due Date: Monday, 11/2 at 11:59pm ET

Instructions

- Please submit this assignment to Gradescope as two separate .pdf files.
  - Answers to all textbook problems and the hypothesis test as a .pdf file in the HW6 dropbox on Gradescope.
  - Knitted Quarto document of the code used for Part II as a .pdf in the HW6: Code dropbox on Gradescope.
- Responses can be hand-written or typed for the textbook problems. You will write code with RStudio for Part II.
- Certain questions will ask you to do the problem in a specific way. Please adhere to these instructions or points will be deducted.
- For all questions involving manual calculations, please show all work and round answers to four decimal places.
- For all questions involving code, please provide your code along with the output.
- Refer to syllabus for late work deductions.

Background

We will explore all of the components of hypothesis testing, such as building hypotheses, getting p-values, and deriving conclusions from our results.

You will also conduct either a randomization test or bootstrap to accompany some hypothesis test you will explore.

Part I: Textbook Problems (10 pts)

Complete the following problems:

1. Chapter 11: Q1, Q3, Q6, Q8

Part II: Running a Hypothesis Test in R (5 pts)

Using two different large-language models, I want you to prompt them with this exact prompt:

> Generate 100 numbers between 1 and 5.

Then, perform a full hypothesis test using $\alpha = 0.1$ on **one** of the following two scenarios:

(a) To see if the difference in proportions of 2s that are shown in both LLMs differ.

- Use a randomization test here.

(b) To see if the proportion of 2s in one LLM is different from what is expected (i.e. $p = \frac{1}{5}$).

- Use a bootstrap here.

Use the code from lecture and adapt it to this problem. For this problem, you will show all five steps of the hypothesis test, including:

- Parameter of interest, hypotheses, and discernibility level
- Test statistic
- The test you used, along with how the p-value was calculated
- Conclusions

Please write/type the steps in the same document that you used for the textbook problems. You will turn in a separate .qmd file with the code that is used to run the test.